Tutorial Questions

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Question 1

A mass of air, at a pressure of 200 kPa and a temperature of 300 K occupies a volume of 3 m^3. Determine;

(a) the density of the air

(b) it's mass

Solution

(a) **Assumption** We are assuming that at these conditions, air is behaving like an ideal gas.

$$P = \rho RT$$
$$\rho = \frac{P}{RT} = \frac{200}{0.287 \times 300}$$
$$\rho = 2.323 \text{ kgm}^{-3}$$

(b) We can find the mass by multiplying the density by the volume.

$$m = \rho \times V$$
$$m = 2.323 \times 3 = 6.97 \text{ kg}$$

Question 2

Calculate the density of air when the absolute pressure and temperature are respectively 140 kPa and $50°C$ and $R = 0.287 \text{ kJ} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$

Solution

Since we are going to assume that the air behaves like an ideal gas, we can use the ideal gas equation to find the density.

$$P = \rho RT$$
$$\rho = \frac{P}{RT} = \frac{140}{0.287 \times (50 + 273)}$$
$$\rho = 1.51 \text{ kgm}^{-3}$$

Question 3

Eight kilometres below the surface of the ocean the presssure is 81.7 MPa. Determine the density of sea-water at this depth if the density at the surface is 1025 kgm^{-3} and the average bulk modulus of elasticity is 2.34 GPa.

Solution

Assumption: • The pressure below the surface of the ocean is the gage pressure, not the absolute pressure. • The density varies with depth, hence the equation $P = \rho gz$ would not be valid. The bulk modulus of elasticity is defined by;

$$\kappa = \rho \left(\frac{\Delta P}{\Delta \rho} \right)_T$$

Making $\Delta\rho$ the subject, we get;

$$\Delta\rho = \rho \left(\frac{\Delta P}{\kappa} \right)$$
$$= 1025 \left(\frac{(81.7 - 0)}{2.34 \times 10^3} \right)$$
$$\Delta\rho = 35.79 \text{ kg} \cdot \text{m}^{-3}$$

Adding the change to our density gives us;

$$\rho_2 = \rho + \Delta\rho$$
$$\implies (1025 + 35.79) \text{ kg} \cdot \text{m}^{-3}$$
$$\rho_2 = 1060.79 \text{ kg} \cdot \text{m}^{-3}$$

Question 4

At an absolute pressure of 101.3 kPa and a temperature of $20°C$ the dynamic viscosity of a certain diatomic gas is 2×10^{-5} Pa · s and its kinematic viscosity is 15 $\text{mm}^2 \cdot \text{s}^{-1}$. Taking the universal gas constant as 8.310 $\text{kJ} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$ and assuming the gas to be perfect, calculate its approximate relative molecular mass.

Solution

Since it is assumed to be an ideal gas, the ideal gas equation will be valid for use here.

$$P = \rho RT$$

Manipulating this to make the molecular mass (M) the subject, we get;

$$P = \rho RT$$
$$P = \frac{\rho R_u T}{M}$$
$$M = \frac{\rho R_u T}{P}$$

We have our pressure, temperature and universal gas constant values already, whar is left is to find the density.
From viscosity, if you divide the dynamic viscosity by the kinematic viscosity, you get the density of the fluid. Applying that gives us;

$$\frac{\mu}{k} = \rho$$
$$\rho = \frac{2 \times 10^{-5}}{15 \times 10^{-6}}$$
$$\rho = 1.333 \text{ kg/m}^3$$

Now we have all the needed parameters, M then becomes;

$$M = \frac{(1.333 \times 8.31 \times (20 + 273))}{101.3}$$
$$M = 32.04 \text{ kg/kmol}$$

Question 5

A hydraulic ram 200 mm in diameter and 1.2 m long moves wholly within a concentric cylinder 200.2 mm in diameter, and the annular clearance is filled with oil of relative density 0.85 and a kinematic viscosity of 400 $\text{mm}^2 \cdot \text{s}^{-1}$. What is the viscous force resisting the motion when the ram moves at 120 mm/s?

Solution

$$F = \frac{\mu \times 2\pi RL \times V}{\ell}$$
$$\implies \frac{(\rho_{H_2O} \times \gamma \times k) \times 2\pi RL \times V}{\ell}$$
$$\implies \frac{(1000 \times 0.85 \times 400 \times 10^{-6}) \times 2\pi \times 0.1 \times 1.2 \times 0.12}{0.0001}$$
$$\therefore F = 307.6 \text{ N}$$

Question 6

The space between two large, flat and parallel walls 25 mm apart is filled with a liquid with dynamic viscosity 0.7 $Pa \cdot s$. Within this space is a thin flat plate 250 mm $\times$ 250 mm is towed at a velocity of 150 mm/s at a distance of 6 mm from one wall, the plate and its movement being parallel to the walls. Assuming linear variations of velocity between the plate and the walls, determine the force exerted by the liquid on the plate.

Solution

$$F = \left(\frac{\mu AV}{\ell}\right) + \left(\frac{\mu AV}{l - \ell}\right)$$

$$F = \left(\frac{0.7 \times (0.25^2) \times 0.15}{0.006}\right) + \left(\frac{0.7 \times (0.25^2) \times 0.15}{0.025 - 0.006}\right)$$

$$F = 1.439 \text{ N}$$

Question 7

A uniform film of oil 0.13 mm thick separates two discs, each of 200 mm diameter, mounted co-axially. Ignoring the edge effects, calculate the torque necessary to rotate one disc relative to the other at a speed of 7 rev/s if the oil has a dynamic viscosity of 0.14 Pa $\cdot$ s.

Solution

$$Differential\ torque,\ dT = \frac{\mu \times (2\pi r dr)(V)}{\ell} \times r$$

$$T = \int_0^R \frac{4\mu\pi^2 r^3 n}{\ell} dr$$

$$T = \frac{\mu\pi^2 r^4 n}{\ell} = \frac{0.14\pi^2 \times 0.1^4 \times 7}{0.00013}$$

$$T = 7.44 \text{ N} \cdot \text{m}$$

Question 8

By how much does the pressure in a cylindrical jet of water 4 mm in diameter exceed the pressure of the surrounding atmosphere if the surface tension of water is 0.073 N $\cdot$ m^{-1}?

Solution

$$\begin{aligned} \Delta p &= \frac{2\sigma_s}{R} \\ \implies & \frac{2 \times 0.073}{0.004} \\ \Delta p &= 36.5 \text{ N} \cdot \text{m}^{-2} \end{aligned}$$

Question 9

What is the approximate capillary rise of water in contact with air (surface tension 0.073 Nm^{-1}) in a clean glass tube 5 mm in diameter?

Solution

The approximate capillary rise in the tube is determined by this function;

$$\begin{aligned} h &= \frac{2\sigma_s \cos\phi}{\rho g R} \\ \implies & \frac{2(0.073)\cos(0)}{1000 \times g \times 0.0025} \\ h &= 5.95 \text{ mm} \end{aligned}$$

Question 10

What is the approximate capillary rise of mercury (relative density 13.56, interfacial tension 0.377 Nm^{-1}, angle of contact approximately 140°) in contact with water in a clean glass tube 6 mm in diameter? (Note: As the mercury moves it displaces water, the density of which is not negligible.)

Solution

$$\begin{aligned} h &= \frac{2\sigma_s \cos\phi}{\rho g R} \\ \implies & \frac{2(0.733)\cos(140)}{((13.56-1) \times 1000) \times g \times 0.003} \\ h &= -1.563 \text{ mm} \end{aligned}$$